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Detection of sleep apnea using deep neural networks and single-lead ECG signals

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ABSTRACT

Sleep apnea causes frequent cessation of breathing during sleep. Feature extraction approaches play a key role in the performance of apnea detection algorithms that use single-lead electrocardiogram signals. Handcrafted features have high computational complexity due to their large dimensions and are usually not robust. To cope with the mentioned problems, in the current paper, an automatic feature extraction method is developed by combining the Convolutional Neural Network (CNN) and Long Short-Term Memory (LSTM) recurrent network. Also, the fully connected layers are utilized to distinguish apnea events from the normal segments. Then, the apnea-hypopnea index (AHI) is applied to discriminate apnea subjects from healthy ones. Finally, in order to assess the usefulness of the proposed method, some experiments are conducted on the publicly accessible Apnea-ECG and UCDDB datasets. The results based on the sensitivity (94.41%), specificity (98.94%), and accuracy (97.21%), indicate that our proposed method provides significant improvements compared to the other sleep apnea detection methods. Our model also achieves an accuracy of 93.70%, sensitivity of 90.69%, and specificity of 95.82% for UCDDB dataset. It can be inferred that using the deep-learning based algorithm for detecting apnea patients would help physicians in making a decision more accurately.

1. Introduction

Safe sleep affects the quality of our life and it is an important mechanism for maintaining mental health. When sleep apnea occurs, the patient's breathing repeatedly stops and starts due to obstruction of the upper airways for at least 10 s during sleep [1,2]. The duration of apnea events is often 10 to 30 s and can occur hundred times during sleep at one night. Sleep apnea can cause complete obstruction (Apnea) or partial (Hypopnoea) obstruction of the upper airways [1,3]. Obstructive sleep apnea (OSA) is the most common type of sleep apnea [4]. In the world, roughly 4% of men and 2% of women suffer from OSA [5]. Failure to diagnose sleep apnea promptly can lead to insomnia, vehicle accidents, and academic under-achievement in children and adolescents.

OSA can cause over-drowsiness (excessive sleepiness) throughout the day, car crashes due to sleep deprivation, mood changes, memory loss, heart disease, and so on [6]. Clinically, sleep apnea is diagnosed using nocturnal polysomnography (PSG) recordings of the patient. The PSG monitors many physiological activities, such as muscle activity (EMG), Electrocardiogram (ECG), eye movements (EOG), Electromyogram (EMG), and Electroencephalogram (EEG) during sleep [7]. To detect sleep apnea at least one night of the PSG recording is required. A PSG typically records at least 12 channels and is very expensive because it contains several costly electrophysiological tests [8,9]. So far, many studies have been performed on the diagnosis of sleep apnea, using a single-lead signals, such as electrocardiogram [10–13], electroencephalogram [14], peripheral oxygen saturation (SpO2) [15], or nasal airflow [16]. Some of the studies such as [17,5,9] proved that the ECG signals can be used to detect respiratory events. Moreover, the results of [18,19] showed that autonomic activity during apnea events and their impact on the cardiovascular system can be measured using the heart rate variability (HRV) signal. Moreover, real-time detection of OSA in patients whose physiological signals are constantly monitored practically is impossible. As a result, developing a low cost computer-assisted OSA screening algorithm is necessary.

To tackle these problems, in literature, two categories of studies including feature engineering-based techniques and deep learning-based methods have been investigated. In feature engineering-based algorithms, the authors used handcrafted features and combined them with different classifiers to improve apnea detection accuracy. For

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example, Atri and Mohebbi extracted useful information from the HRV and ECG derived respiratory (EDR) signals using the higher-order spectrum based technique. Then, they classified the feature vectors using the least-square support vector machine (LS-SVM) classifier [20]. Tripathy et al. utilized the intrinsic band functions algorithm to decompose both the HRV and EDR signals into different subbands. Then, they calculated different features and fed them into the kernel extreme learning machine classifier [21]. Nishad et al. used the centered correntropies and tunable-Q wavelet transform (TQWT) based filter-bank to classify one-minute ECG segments [22]. Sharma et al. extracted non-linear features from different sub-bands of ECG signal, using the biorthogonal antisymmetric wavelet filter bank, and classified them into healthy and OSA groups using the LS-SVM [23]. Li et al. introduced a novel OSA detection index using the sliding trend fuzzy approximate entropy (SlTr-fApEn), empirical mode decomposition (EMD) method, and the HRV signal [24]. Zarei and Asl presented a novel algorithm based on the discrete wavelet transform and non-linear features to detect apnea events using ECG signals [25]. In another study, they introduced a new automatic OSA detection technique to classify one-minute ECG segments. They extracted non-linear features such as fuzzy entropy from HRV and EDR signals [2]. Gonzalez et al. calculated different features, namely Cepstrum coefficients and detrended fluctuation analysis from the HRV signal, and fed them into different classifiers for diagnosing sleep apnea [26]. The authors in [27] implemented a new apnea recognition framework based on the autoregressive modeling, spectral autocorrelation function, and ECG signals. They employed the random forest model to classify extracted features. Faal et al. introduced a new apnea detection method using ECG signals and statistical modeling. They applied different classifiers on the features calculated from the time domain [28]. Fatimah et al. extracted different handcrafted features in the frequency domain and investigated the capability of the Fourier decomposition approach in OSA detection [29]. Mostafa et al. [30] performed two different techniques to choose the best subset of features. In their study, the selected features were tested with different classification methods.

Also, some other studies have been focused on deep learning-based techniques. For example, Li et al. worked on the recognition of the OSA segments using the hidden Markov model (HMM), deep neural network, and ECG signals. They employed the sparse auto-encoder to extract features and then classified them using the SVM and artificial neural networks (ANN) [31]. Choi et al. developed an automatic apnea detection approach using convolutional neural networks (CNN) and PSG signals [32]. The authors in [33] investigated the effectiveness of different deep learning structures in apnea detection application. They employed the CNN and RNN based structures to automatically recognize apnea events using ECG signals. Dey et al. introduced a supervised apnea detection approach using the CNN structure and ECG signal [34]. Novak et al. used the HRV signal and LSTM network to diagnose apnea patients [35]. Pathinarupothi et al. presented a new apnea detection framework by employing the LSTM-RNN structure on instantaneous heart rates [36]. It seems that two main problems remain unsolved in many studies related to this field of study. Firstly, the performance of previous OSA recognition methods strongly depends on the feature calculation approaches. Furthermore, the performance of the traditional handicraft feature extraction methods depends on the experience of specialists and their prior knowledge of physiological signals. Secondly, notwithstanding the researchers applied different classification models, they have not focused on improving the performance of the classifiers [31]. Therefore, developing algorithms with less dependency on feature engineering is necessary to expand the application range of computer-aided techniques.

To deal with these problems, an automatic feature extraction and classification technique is developed using deep neural networks. One of the goals of deep learning algorithms is to achieve feature learning in an unsupervised manner. Deep learning is a fully trainable system that can start with the raw data, extract the most informative features automatically, and recognize objects in its output by the neural network. Although attempts have been made previously on the detection of the OSA via CNN and Long Short-Term Memory (LSTM) networks [33], the algorithm presented here is different from the existing techniques. Erdenebayar et al. assessed the effectiveness of the one dimensional (1D) CNN, two dimensional (2D) CNN, and LSTM networks individually [33]. The authors in [33] excluded half of the existing dataset (the withheld set). In the current work, both of the released and withheld sets have been used. This study has combined the 2D-CNN and LSTM networks to extract more informative features from single-lead ECG signals at the same time. There are two main reasons for using this architecture:

- Extracting deep features and improving the classification accuracy; If diverse layers are used, they can extract more deep features. Also, the CNN-LSTM is a common architecture [37] for time-series applications.
- Decreasing the computational complexity; High computational complexity is encountered when only the LSTM layers are used.

The rest of the current study is continued as follows. All the materials and methods are explained in Section 2. The results are given in Section 3. Finally, discussion and conclusion are presented in Sections 4 and 5, respectively.

2. Materials and methods

2.1. Proposed framework

In the present study, we developed an automatic OSA classification approach based on deep neural networks and ECG signals. The main idea of this study is developing a fully automatic (or an end-to-end) deep learning-based OSA detection method. In the current study, all the feature extraction and classification processes have been done using different types of deep learning layers. Also, in order to eliminate the need of additional tuning step, tuning the optimal value of different parameters of classifiers, deep learning is used instead of conventional classifiers. In the pre-processing step, the noisy segments are detected and eliminated by the weight calculation algorithm. Then, the most prominent features are automatically extracted using an end-to-end deep learning structure. The schematic design of the proposed technique in this study is demonstrated in Fig. 1. As it is represented in this figure, in our proposed method, the CNN and LSTM networks are used to automatically calculate the spatial and temporal features, respectively. Also, the fully connected layers are applied to identify apnea events. At the end of this stage, the AHI index is employed to diagnose apnea patients.

2.2. Datasets

2.2.1. Apnea-ECG dataset

The Apnea-ECG dataset contains divided into the released and the withheld sets. Each of the released and withheld sets contains 35 recordings. Expert physicians have labeled each of the one-minute ECG segments as the normal or apnea ones. The length of the recordings varies between 401 min to nearly 578 min. The age of the subjects (57 men, 13 women) ranged from 27 to 63 years and weights from 53 to 135 kg. The total number of released-set segments is 17045 one-minute segments. Also, the withheld-set consists of 6550 apnea, and 10718 normal segments. In the present study, the subjects with $AHI < 5$ were considered as healthy subjects otherwise, they were defined as OSA subjects. Furthermore demographic information and physiological parameters of subjects are summarized in Table 1. More details can be found in [38].

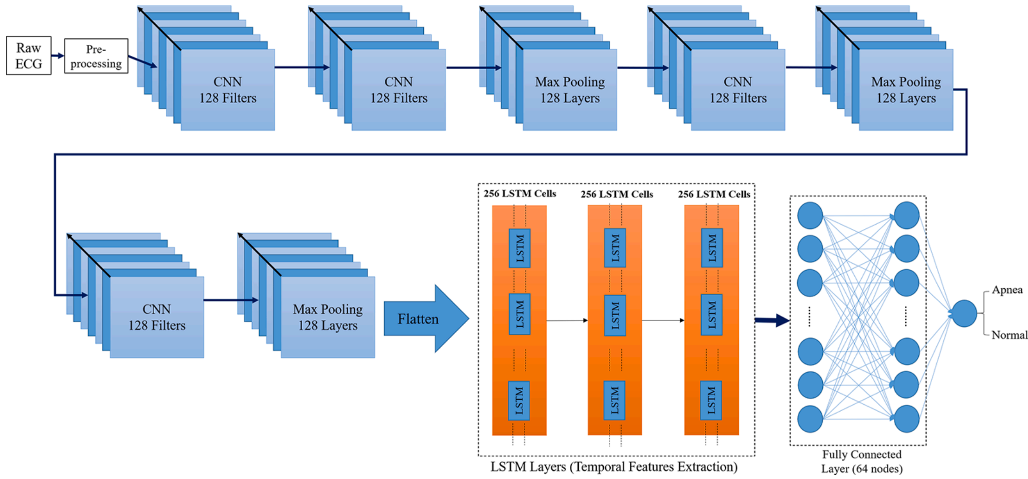

Fig. 1. Flowchart of proposed sleep apnea detection algorithm based on CNN and LSTM networks.

Table 1
Demographic information and physiological parameters of poupulation. BMI: body mass index, AI: Apnea index, HI: hypopneas index, AHI: apnea-hypopnea index, TRT: total recording time, AT: Apnea time, Non-AT: Non-Apnea time.

Parameters	Released Set		Withheld Set	
	Normal (*mean* ± *std*)	Apnea (*mean* ± *std*)	Normal (*mean* ± *std*)	Apnea (*mean* ± *std*)
Age (years)	35.5 ± 6.32	52.17 ± 6.09	32.37 ± 6.09	49.08 ± 9.46
BMI (kg/m^2)	21.41 ± 2.20	31.41 ± 5.34	21.10 ± 1.22	31.22 ± 5.27
Height (cm)	177.5 ± 6.17	176.17 ± 3.23	172.27 ± 9.61	176.33 ± 4.18
Weight (kg)	67.58 ± 8.78	97.43 ± 16.46	62.73 ± 7.16	97.13 ± 16.63
AI (events/h)	0.68 ± 0.2	32.26 ± 20.89	0.06 ± 0.12	32.78 ± 24.85
HI (events/h)	0.03 ± 0.08	11.27 ± 12.90	0.0 ± 0.0	7.23 ± 6.50
AHI (events/h)	0.10 ± 0.21	43.53 ± 21.87	0.06 ± 0.12	40.01 ± 25.42
TRT (min)	472.5 ± 29.38	498.04 ± 30.23	461.73 ± 25.83	509.33 ± 20.68
AT (min)	3.42 ± 5.66	281.43 ± 136.22	2.27 ± 3.50	271.96 ± 144.61
Non-AT (min)	469.08 ± 30.64	216.61 ± 124.67	459.45 ± 25.71	237.38 ± 141.07

2.2.2. St. Vincent's University Hospital/ University College Dublin dataset (UCDDB)

In the current work, this dataset has also been used to validate the performance of the proposed OSA detection method. There are 25 full overnight polysomnograms with simultaneous three-channel Holter ECG in this dataset and it was recorded from adult subjects with suspected sleep-disordered breathing. The weights and ages of the subjects vary between 59.8–128.6 kg and 28–68 years, respectively. Also, the AHI values are in the range of 1.7–90.9. Frequency sampling of all the ECG recordings is 128 *Hz*. For more details the readers are referred to [39].

2.3. Pre-processing

At first, the Chebyshev bandpass filter with the cutoff frequencies of 0.5 *Hz* and 48 *Hz* is designed to suppress the baseline wandering and power line interference noises. The ECG signals are then divided into one-minute (for Apnea-ECG dataset) and 30-s (for UCDDB dataset) segments. Afterward, the noisy segments are eliminated using an automatic weight calculation algorithm. This algorithm measures the similarity of the autocorrelation functions (ACF) of the segments with the time lags of 60 s. At the output of this process, a similarity matrix is obtained, which is a symmetric matrix. At the next stage, the weights vector will be obtained by summing each of the columns of the similarity matrix. The threshold value of 0.8 is used to recognize and eliminate the noisy segments. This threshold value is selected by try and error after conducting several experiments. In the this stage, less than 5% of the segments (1717 segments of Apnea-ECG dataset) were eliminated from the experimental analysis. Further details can be found in [25].

2.4. Convolutional neural network

Convolutional neural networks (CNN) are one of the most common and highly efficient techniques that are widely used in various signal and image processing applications. In recent years, this method has attracted much attention in analyzing physiological events, such as sleep stage scoring [40], sleep apnea detection [33], or seizure detection [41]. CNN networks consist of neurons with weights and configurable biases. Each neuron receives a part of the ECG segment as an input and then calculates the multiplication of the weights in the input segment values. The output of this process is converted to class labels using a non-linear activation function. Also, the network provides a differentiable score function, with inputs on one side and scores on the other side.

In general, CNN networks consist of three parts: convolutional, pooling, and fully connected parts. The convolutional part consists of filters called kernels and convolve with the ECG segments. This process produces the feature vector. Afterward, the max-pooling operation is employed to the feature vector. In this step, the values of each group of neighbors are combined with a summarization function. The summarization function performs like a low-pass operator, and common choices for it are either maximal or average. The max-pooling layer reduces the computation burden and prevents overfitting by selecting the largest value of each feature. The fully connected part often is used as the last part of the CNN networks and classify the output of the max-pooling layer (the feature vectors).

2.5. Recurrent neural network

A Recurrent neural network (RNN) commonly measures the correlation between the values of signals during the time. In many practical situations, such as signal processing, there is a meaningful dependency between different sequences. The RNN networks have a memory unit that can process sequences of the input signal [42]. The RNNs-based structures can extract temporal features and measure temporal dependencies.

2.6. Long Short Term Memory

Theoretically, RNNs show excellent performance in maintaining the correlation of input signals whereas in practical application, they encounter a lot of restrictions to measure these time dependencies. To cope with these challenges, the Long Short Term Memory (LSTM) structure is proposed which is an extended version of RNNs [42]. The LSTM structures capture signal variations using long and short-term memory. An example of a common LSTM cell is depicted in Fig. 2. As demonstrated in this figure, each cell composed of three gates; an input gate, an output gate, and a forget gate. Each of the mentioned gates can be defined as follows:

Forget gate:

$$\mathbf{f_t} = \sigma_g\left(\mathbf{W_f x_t} + \mathbf{U_f h_{t-1}} + b_f\right) \tag{1}$$

Input gate:

$$\mathbf{i_t} = \sigma_g(\mathbf{W_i x_t} + \mathbf{U_i h_{t-1}} + b_i) \tag{2}$$

Cell state update:

$$\mathbf{C_t} = \mathbf{f_t} \odot \mathbf{C_{t-1}} + \mathbf{i_t} \odot \sigma_c(\mathbf{W_c x_t} + \mathbf{U_c h_{t-1}} + b_c) \tag{3}$$

Output gate:

$$\mathbf{O_t} = \sigma_g(\mathbf{W_o x_t} + \mathbf{U_o h_{t-1}} + b_o) \tag{4}$$

Output:

$$\mathbf{h_t} = \mathbf{O_t} \odot \sigma_h(\mathbf{C_t}) \tag{5}$$

where both of the σ_h and σ_c are hyperbolic tangent functions. It is noteworthy that weight matrices, $\mathbf{W}$ and $\mathbf{U}$, are optimized during training. In this expression, the output and state of the previous LSTM cell are defined using the $\mathbf{C_{t-1}}$, and $\mathbf{h_{t-1}}$, respectively. The bias vector and input vector are denoted with b and $\mathbf{x_t}$, respectively. Also, σ_g is the sigmoid function. Finally, the operator $\odot$ denotes the Hadamard product. In an LSTM cell, weights can be updated using the $\mathbf{C_{t-1}}$ and $\mathbf{i_t}$. The LSTM cell can capture the long interval dependency of the input time series because it uses the gating mechanism [43].

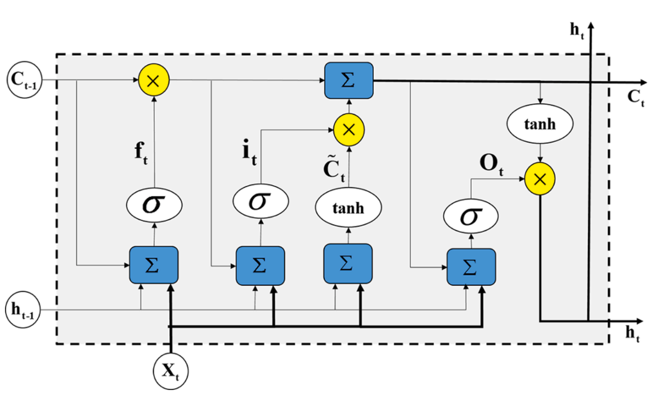

Fig. 2. A typical architecture of a LSTM cell. A LSTM block typically has a memory cell, input gate ($\mathbf{i_t}$), output gate ($\mathbf{O_t}$), and a forget gate ($\mathbf{f_t}$) in addition to the hidden state ($\mathbf{h_t}$) in traditional RNNs.

2.7. The proposed model for OSA recognition

In the literature, most of the researchers introduced handcraft feature extraction solution. In these methods, features are extracted manually and their performance is highly dependent on the experience of specialists. To cope with the mentioned problem, an automatic end-to-end deep learning structure was utilized. Moreover, the proposed automatic structure can extract the most informative features and increase the classification accuracy. As indicated in Fig. 1, our proposed model consists of two main parts. The first part is the feature extraction contains four 2D CNN and three LSTM layers. These layers are utilized to calculate the most prominent features of the ECG signals. In the second part, the classification phase, the fully connected layer is applied to the feature vector to discriminate apnea segments from normal ones. Also, to keep away from the overfitting problem, the dropout layers were applied. This method is a regularization approach for deep networks. According to this solution, at each epoch, some of the input units randomly were set to zero during the training time [44]. Mathematically, the output of the LSTM cell can be calculated as follows:

$$\begin{pmatrix} \mathbf{f} \\ \mathbf{o} \end{pmatrix} = \begin{pmatrix} sigmoid \\ sigmoid \\ sigmoid \\ \tanh \end{pmatrix} \left[\begin{pmatrix} \mathbf{x_t} \\ \mathbf{h_{t-1}} \end{pmatrix} . \mathbf{W} \right] \tag{6}$$

where $\mathbf{W}$ demonstrates the weight matrix which its dimension is $2k \times 4k$ and k is the dimension of the input signal. Also, the dropout variation is formulated as follows:

$$\begin{pmatrix} \mathbf{f} \\ \mathbf{o} \end{pmatrix} = \begin{pmatrix} sigmoid \\ sigmoid \\ sigmoid \\ \tanh \end{pmatrix} \left[\begin{pmatrix} \mathbf{x_t} \times \mathbf{z_x} \\ \mathbf{h_{t-1}} \times \mathbf{z_h} \end{pmatrix} . \mathbf{W} \right] \tag{7}$$

where $\mathbf{z_x}$ and $\mathbf{z_h}$ are the random filter masks which drop some weights in each iteration. In order to tune hyperparameters, the grid search method has been utilized to achieve an optimal structure with the highest accuracy and lowest computational complexity. To this end, the number of LSTM and CNN layers was changed from 1 to 6 layers. Different values including 32, 64, 128, and 256 were used for the number of cells and filters in LSTM and CNN layers, respectively. Also, other parameters such as learning rate, activation functions, and the kernel size were determined empirically. Finally, some of the optimal parameters of our proposed structure are summarized in Table 2.

2.8. Experimental setting

As mentioned before, in this study, an automatic OSA detection technique was presented using CNN and LSTM networks. In the training phase, the effect of parameters of our proposed deep structure were investigated using the released-set (Fig. 3). In order to extract spatial features, a 2D CNN model was employed. Therefore, one-dimensional

Table 2
The learning parameters of the proposed deep learning structure.

Task	Layers	Leaning Parameters
Feature Extraction	CNN layers	number of filters = 128, kernel size = 2 × 2, activation = ReLU
	Max-Pooling	Pool size = 2 × 2
	LSTM layers	number of cells = 256
Classifier	Optimizer	Adam (learning rate = 0.0005, decay = 1e-6)
	Loss function	Binary cross-entropy
	Activation functions	ReLU and Sigmoid (lastest layer)
	Number of units	64

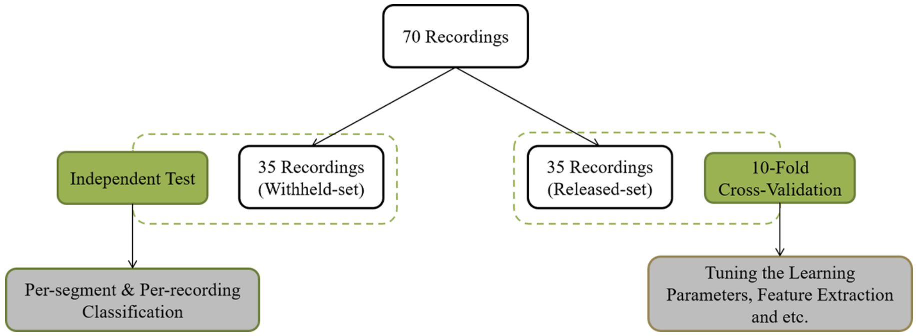

Fig. 3. Dataset division for per segment and per-recording classification tasks.

ECG segments were converted to two-dimensional segments. Afterward, temporal features were calculated using the LSTM networks. To remove any simple dependencies between the neurons, i.e., to cope with the overfitting problem, the dropout function with the probability of 0.2 was applied after some layers. This method randomly sets 20 percent of input units to zero during the training time. The optimal parameters of feature extraction and classifier layers were presented in Table 2. As detailed in Table 2, the Adam optimizer and binary cross-entropy loss function were applied to optimize the values of various parameters [45]. The performance of the proposed model (accuracy and loss) along 200 epochs was depicted in Fig. 4. In this figure, the number of epochs was changed from 1 to 200 epochs. As demonstrated in Fig. 4 (C), although the accuracy increases (or the loss value decreases in Fig. 4 (D)) with increasing the number of epochs in the training time, in the test phase the accuracy (Fig. 4 (A)) and loss function (Fig. 4 (B)) converge to a constant value after 50 epochs. Finally, the details of the CNN–LSTM based model are presented in Table 3. Our model was implemented based on the Tensorflow framework on GeForce GTX 1080Ti graphic card with 12 GB memory.

Different criteria such as precision, specificity, F1-Score, sensitivity, and accuracy were used for performance evaluation of our proposed technique.

3. Results

In the this work, a novel end-to-end learning structure has proposed to diagnose OSA. The model performance was evaluated through different aspects including classifier parameter tuning, per segment classification using different criteria, and subject-by-subject (per-recording) OSA classification accuracy.

3.1. Tuning classifier parameters

In the present section, the effect of setting various classifier

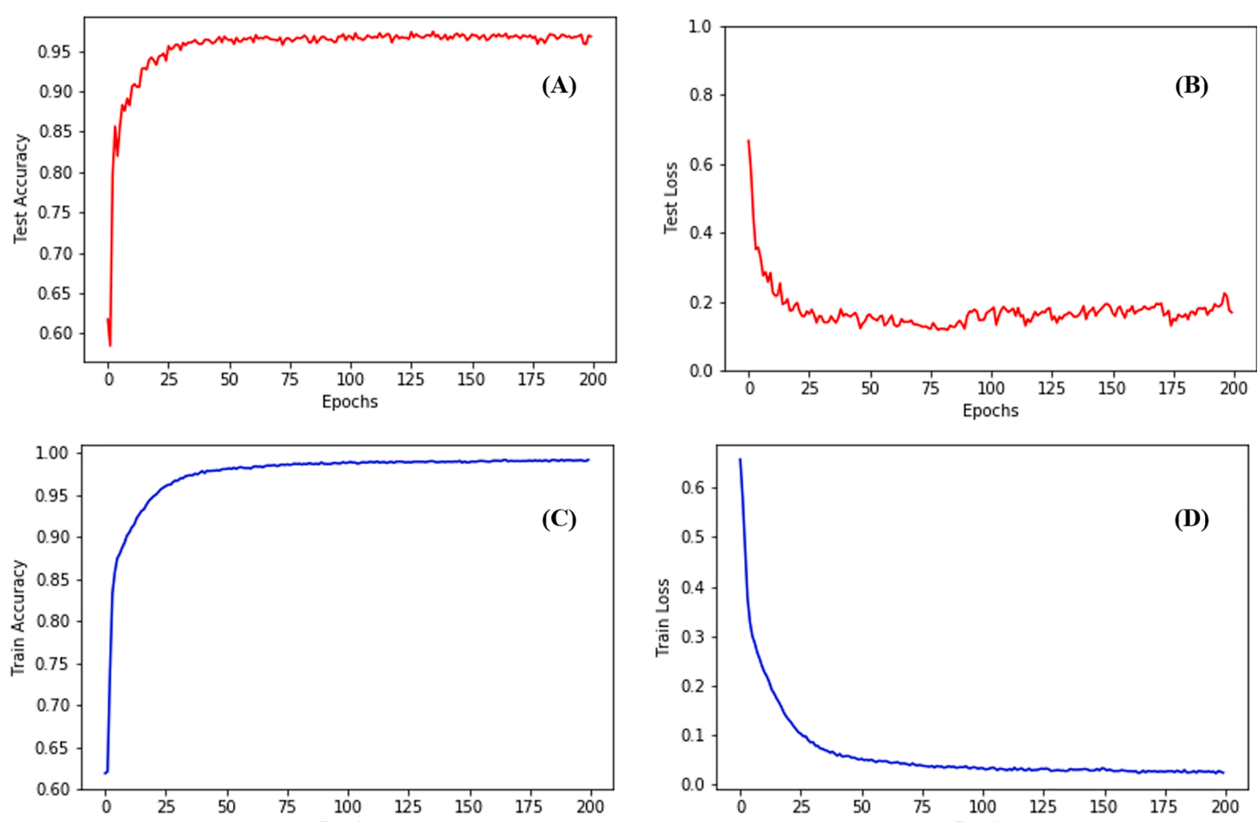

Fig. 4. Performance of the proposed network along 200 epochs. (A) and (B) are the accuracy and loss values of the test phase, respectively. Also, (C) and (D) are the accuracy and loss values of the training phase, respectively.

Table 3
The detailed specifications of CNN-LSTM layers.

Layer	Type	Number of filter/cell/unit	Kernel size	Activation function
1	CNN	128	2 × 2	ReLu
2	CNN	128	2 × 2	ReLu
3	MAX-Polling	–	2 × 2	–
4	Dropout	–	–	–
5	CNN	128	2 × 2	ReLu
6	MAX-Polling	–	2 × 2	–
7	Dropout	–	–	–
8	CNN	128	2 × 2	ReLu
9	MAX-Polling	–	–	–
10	Dropout	–	–	–
11	Flatten	–	–	–
12	LSTM	256	–	–
13	LSTM	256	–	–
14	LSTM	256	–	–
15	Dropout	–	–	–
16	Dense	64	–	ReLu
17	Dense	1	–	Sigmoid

parameters, the number of neurons and layers, on the accuracy of the proposed model is assessed. As reported in Table 4, four different DNN structures with different parameters were examined. The results show that the model becomes overfit by increasing hidden layers. So, the DNN2 which has one hidden layer with 64 neurons was employed for the classification task.

3.2. Per-segment classification

The withheld-set was utilized to assess the effectiveness of the introduced model. In this step, the ECG segments fed into the combination of CNN and LSTM networks to automatically extract features. Then, the extracted features were fed into fully connected layers to classify one minute ECG segments. The result of the optimized CNN-LSTM structure for per segment classification were reported in Table 5. In accordance with this table, our developed CNN-LSTM structure provides the F1-score of 0.96, PPV of 94.42%, accuracy of 97.21%, sensitivity of 94.41%, and specificity of 98.94%. It is obvious that the presented technique can improve classification performance using the combination of CNN and LSTM networks (See Table 6).

For automatic classification of one-minute ECG segments, various deep learning structures including CNN, LSTM and CNN-LSTM were examined to find the best model. Table 11 summarizes the results of the designed methods.

Different deep learning structures including CNN, LSTM, and CNN-LSTM were examined to find the best model for the automatic classification of one-minute ECG segments. The results of the designed techniques are summarized in Table 11. According to Table 11, in the test set, the highest sensitivity (95.35%) is obtained using the CNN model. The results indicated that the designed LSTM model has the worst performance among the other models. The CNN-LSTM model outperforms the other designed models by providing an accuracy of 97.21%, sensitivity of 94.41%, and specificity of 98.94%. This is because the CNN and LSTM networks extract the spatial and temporal features at the same time. In per segment classification, the highest accuracy value of previous methods on the Apnea-ECG dataset is listed in Table 7. The results show that (Table 7) our developed OSA detection method performs better than existing studies in terms of accuracy, sensitivity and specificity.

Table 4
The learning parameters of different layers of classifier.

	DNN1	DNN2	DNN3	DNN4
Hidden layer 1	32 unit	64 unit	32 unit	64 unit
Hidden layer 2	–	–	32 unit	64 unit
Accuracy	96.41 %	97.21 %	96.16 %	96.73 %

Statistical power or the power of a hypothesis test is the probability that the test correctly rejects the null hypothesis. That is the probability of a true positive result. It is only useful when the null hypothesis is rejected. The higher the statistical power for a given experiment, the lower the probability of making a Type II (false negative) error. Experimental results with too low statistical power will lead to invalid conclusions about the meaning of the results [60]. To assess the statistical power of the presented method, the model generated by data of 35 randomly selected recordings was applied to the data of remaining recordings (35 recordings) as test data. This procedure was repeated 10 times and the results are presented in Table 8. Also, Fig. 5 illustrates the statistical power of the proposed method for a range of accuracy. As indicated in Fig. 5, the statistical power of our proposed method for a true mean (the average accuracy obtained by previous runs, i.e., 97.21%) is one. Therefore, we can see that the proposed technique has a great performance in terms of statistical power analysis.

Execution times of the proposed algorithm on the Apnea-ECG and UCDDB datasets are reported in Table 9. According to this table, it is obvious the computational cost of the proposed OSA screening scheme is also significantly low.

3.3. Per-recording classification

In the next step, the AHI index was utilized to separate the healthy subjects from the OSA patients. This criterion is defined as follows:

$$AHI_{Real} = \frac{Number \text{ of apnea segments}}{Total \text{ sleep time (in minutes)}} \times 60. \tag{8}$$

According to this criterion, the subjects with AHI more than five were classified as apneic patients. Also, the cases who have the AHI value less than 5 were considered as healthy subjects. The performance of the presented technique was assessed using the withheld-set which contains 35 recordings. A summary of the per-recording OSA detection performances of various researches along with the proposed method was listed in Table 10. In accordance with this table, our proposed technique outperforms most of the other studies by classifying all subjects correctly (an accuracy of 100%). Also, the mean absolute error (MAE) metric was applied to better evaluate the pre-recording classification performance. This metric measures the gap between the estimated AHI and real AHI and can be calculated as follows:

$$\text{MAE} = \frac{1}{N}\sum_{i=1}^{N}\left|\text{Estimated}^{i}_{AHI} - \text{Real}^{i}_{AHI}\right| \tag{9}$$

where N is the number of recordings. Our proposed deep learning-based method provides the MAE of 2.93 for per-recording OSA detection.

4. Discussion

In the present paper, a novel unsupervised feature extraction framework has been developed to recognize OSA patients using CNN and LSTM networks. In the proposed method, the spatial and temporal features were automatically extracted using the 2D-CNN and LSTM, respectively. Moreover, a fully connected layer was adopted to classify apnea events and normal segments. As a result, a fully automatic OSA detection algorithm was presented.

4.1. Learning representation and temporal dependency

As shown in Table 7, most of the presented OSA detection methods are developed based on the handcrafted features. The results showed that feature extraction techniques play a key role in the performance of

Table 5
The result of the optimized CNN-LSTM structure for per segment classification.

Approach	Criteria								
	TP	TN	FP	FN	Sens (%)	Spec (%)	Acc (%)	F1-score	PPV (%)
CNN-LSTM	6104	10341	361	110	94.41	98.94	97.21	0.96	94.42

Table 6
Performance evaluation of the deep learning approaches on minute-by-minute classification.

Approach	Criteria		
	Sens (%)	Spec (%)	Acc (%)
CNN	**95.35**	96.03	95.77
LSTM	89.70	95.58	93.33
CNN-LSTM	94.41	**98.94**	**97.21**

apnea detection methods. The performance of traditional hand-crafted feature extraction methods is heavily dependent on the experts' knowledge from different physiological signals and exclusive subjects' characteristics. Further, some studies that used automatic feature extraction methods, have just considered time domain correlation in signals and they have not conducted spatial dependencies in ECG signals. In the current study, a new deep learning-based technique was employed to capture the spatial and temporal dependencies using 2D-CNN and LSTM networks, respectively. The reported results in Tables 7 and 10 imply that the proposed model provided outperformance in comparison with other methods.

4.2. Comparison and summary

In the present section, we compared the performance of our introduced technique with approaches that were conducted on the Physionet Apnea-ECG dataset. The comparison of per segment classification performance results are summarized in Table 7. As indicated in this table, our developed approach using Deep Learning can provide an accuracy of 97.21%, sensitivity of 94.41%, and specificity of 98.94%. The results showed that our developed OSA detection technique has a better performance compared to previous algorithms. The main drawback found in most of the previous studies is that their performance strongly depends on feature extraction methods. Feature calculation and feature engineering are the most important and time-consuming parts of the machine learning algorithms.Feature engineering based techniques try to improve performance by mapping high-dimensional training data to low-dimensional feature space. Whereas, in deep learning algorithms, feature learning is done in an unsupervised and automatic manner. To accomplish these objectives, in this study, the 2D-CNN and LSTM networks were applied to automatically extract the spatial and temporal features, respectively. Then, a fully connected layer was applied to identify apnea events.

In Table 10, we compared per-recording classification performance of our developed technique with the OSA detection algorithms using the same dataset. As reported in Table 10, in per-recording classification phase, our deep learning-based algorithm detected all subjects correctly and provided good results (an accuracy of 100% and specificity of 100%). Based on the obtained results, it can be concluded that the presented OSA recognition algorithm outperforms most recent studies. Moreover, in Table 11, the performance of our developed deep structure was compared with previous deep learning-based studies. According to this table, the proposed deep learning method showed better classification performance compared to other techniques.

Also, some other studies such as [6,17,31] have obtained remarkable outcomes. Although high classification results have been obtained in [6], they have excluded 20 recordings with low-quality signals during per-recording classification. Their method may provide low

Table 7
Comparison of segment-by-segment based classification results between our method and previous works on the Physionet Apnea-ECG database.

Reference	Method	Sens (%)	Spec (%)	Acc (%)	Signal
[13]	HRV analysis in the frequency domain	76.7	88.4	86.2	Lead-II ECG
[1]	RQA Statistics and SVM	86.37	83.47	85.26	Lead-II ECG
[25]	Entropy-based features, wavelet transform and SVM	91.74	93.75	92.98	Lead-II ECG
[17]	PCA of QRS complex and HRV features	84.71	84.69	84.74	Lead-II ECG
[46]	HRV, Hermite basis functions, SVM and LS-SVM	79.5	89.6	84.7	Lead-II ECG
[11]	QRS detection, HRV, and EDR-based features + HMM	82.6	88.4	86.2	Lead-II ECG
[26]	Filterbank, Cepstrum, DFA and QDA	81.45	86.82	84.76	Lead-II ECG
[47]	TQWT and RUSBoost	87.58	91.49	88.88	Lead-II ECG
[31]	DNN + HMM	88.9	88.4	83.8	Lead-II ECG
[21]	HRV, EDR, FDM, and KELM	78.02	74.64	76.37	Lead-II ECG
[22]	TQWT, centered correntropys, and random forest	90.95	93.91	92.78	Lead-II ECG
[23]	BAWFB, Entropy-based features, LS-SVM	90.87	88.88	90.11	Lead-II ECG
[2]	HRV, EDR, Non-linear features, GentleBoost classifier	91.52	94.36	93.26	Lead-II ECG
[48]	NIG parameters + TQWT + AdaBoost	81.99	90.72	87.33	Lead-II ECG
[49]	Statistical Features and ELM	85.20	82.79	83.77	Lead-II ECG
[50]	statistical and spectral features + bootstrap aggregating	84.14	86.83	85.97	Lead-II ECG
[51]	Statistical Features and ELM	–	–	83.77	Lead-II ECG
[52]	Statistical Features and ELM	–	–	83.77	Lead-II ECG
[53]	Statistical and Spectral Features + RUSBoost	–	–	85.37	Lead-II ECG
[54]	10-s ECG segment + CNN	96.1	96.2	96.1	Lead-II ECG
[55]	Fused images + CNN	92.3	92.6	92.4	Lead-II ECG
[56]	RR interval + Multiscale Deep Neural Network	89.8	89.1	89.4	Lead-II ECG
[57]	ECG signal + 1D Deep CNN model	81.1	90.0	87.9	Lead-II ECG
[58]	RR interval + FSSAE + Softmax-HMM	86.2	84.4	85.1	Lead-II ECG
[59]	ECG signal, EMD, CWT, and SCNN	94.30	94.51	94.30	Lead-II ECG
Proposed Method	2D-CNN + LSTM networks	**94.41**	**98.94**	**97.21**	Lead-II ECG

performances in the practical conditions in which physiological signals have noisy nature. Some other studies, such as [17], have reached 100% accuracy, but their method is not completely automatic. They have used

Table 8
The results of 10 performances on the test data (data of 35 recordings).

Run	Sens (%)	Spec (%)	Acc (%)
1^{th}	97.32%	97.59%	97.11%
2^{th}	96.57%	96.12%	96.91%
3^{th}	95.51%	96.97%	94.42%
4^{th}	95.95%	97.32%	94.93%
5^{th}	94.11%	90.76%	96.61%
6^{th}	92.80%	83.92%	97.39%
7^{th}	93.08%	90.91%	95.44%
8^{th}	95.70%	95.47%	95.87%
9^{th}	97.33%	97.80%	96.99%
10^{th}	96.88%	97.86%	97.90%
Mean	95.53%	94.47%	96.36%
Std	1.6673%	4.3376%	1.0794%

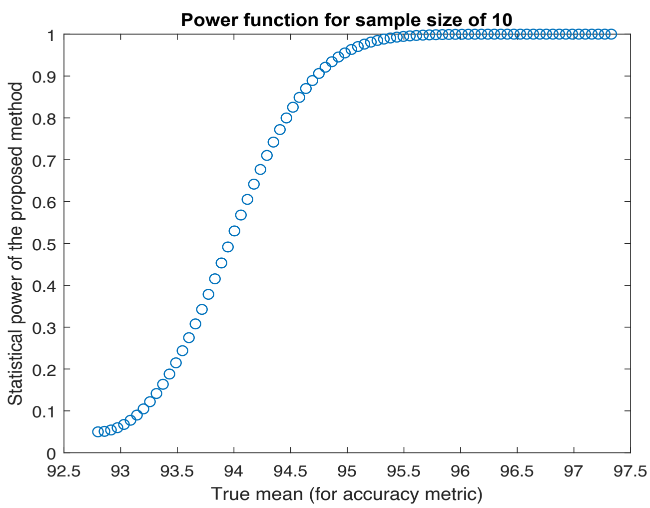

Fig. 5. Statistical power of the proposed method based on accuracy and sample size of 10.

Table 9
Execution time of the proposed algorithm on the Apnea-ECG and UCDDB datasets.

Dataset	Time (second)
One-minute ECG segment with 6000 samples from the Apnea-ECG database	0.0684
30-s ECG segment with 3840 samples from the UCDDB database	0.0353

handcrafted features for classification. However, our proposed algorithm obtained the same results (for per-recording classification) by employing the LSTM and 2D-CNN networks. According to the results presented in Tables 7 and 10, the developed approach has a better performance than the recently presented methods.

The authors in [56] proposed a novel OSA detection technique using RR intervals, multiscale dilation attention 1-D convolutional neural network (MSDA-1DCNN) and a weighted-loss time-dependent (WLTD) classification model. Also, in reference [61], the authors have developed a new OSA recognition system using the RR intervals, CNN, and LeNet-5 networks. There are a lot of differences between our study and the mentioned references. In our proposed method, a fully automatic OSA detection technique has been developed using the ECG signals, 2-D CNN, and LSTM structures. Although all the mentioned studies (the proposed method, [56,61]) are based on CNN, there are many differences in the

Table 10
Comparison of subject-by-subject based classification results between our method and previous works on the Physionet Apnea-ECG database.

Reference	Method	Acc (%)	Sens (%)	Spec (%)	Signals
[56]	RR interval + Multiscale Deep Neural Network	100	100	100	Lead-II ECG
[61]	RR interval + LeNet-5	97.1	100	91.7	Lead-II ECG
[57]	ECG signal + 1D Deep CNN model	97.1	95.7	100	Lead-II ECG
[58]	RR interval + FSSAE + Softmax-HMM	97.1	95.7	100	Lead-II ECG
[59]	ECG signal, EMD, CWT, and SCNN	100	100	100	Lead-II ECG
[17]	PCA of QRS complex and HRV features	100	100	100	Lead-II ECG
[9]	Handcrafted features + machine learning	95.7	100	87	Lead-II ECG
[46]	HRV, Hermite basis functions, SVM and LS-SVM	97.14	95.8	100	Lead-II ECG
[11]	QRS detection, HRV, and EDR-based features + HMM	97.1	95.8	100	Lead-II ECG
[31]	DNN + HMM	100	100	100	Lead-II ECG
[25]	Entropy-based features, wavelet transform and SVM	95.71	95.83	95.66	Lead-II ECG
This work	**ECG signal + CNN-LSTM**	**100**	**100**	**100**	Lead-II ECG

Table 11
Performance comparison with previous deep learning-based studies.

Study	Acc (%)	Sens (%)	Spec (%)	Subject	Method
[36]	98	–	–	35	LSTM
[62]	97.8	–	–	35	LSTM
[34]	98.9	97.8	99.2	35	CNN
[32]	96.6	81.1	98.5	179	CNN
[33]	99	99	99	86	GRU
[63]	97.14	95.8	100	35	LSTM
[35]	82.1	85.5	80.1	33	LSTM
[61]	97.1	100	91.7	–	LeNet-5
[56]	100	100	100	70	MDNN
Our method	100	100	100	70	LSTM + CNN

proposed structures. For example, a combination of 2D-CNN and LSTM networks has been used in our proposed method to diagnose OSA events. However, in the mentioned references ([56,61]), the 1-D CNN network has been applied to detect apnea events. In summary, we can see that the optimization process, feature extraction process, and classification process are different between the mentioned studies.

Furthermore, the performance of the proposed deep learning-based OSA detection method was validated on UCDDB database. In our experiments we found that decreasing the number of the ECG segments in the input leads to decreasing the performance of the deep learning-based techniques. Because the number of segments in the UCDDB database is less than the Apnea-ECG database, the 80–20 ratio was used to evaluate the performance of the proposed method on the UCDDB database. Consequently, to evaluate the performance of our proposed technique 80% of recordings were randomly selected for training the model and the remaining (20% of the recordings) were used for evaluating the performance. This process was repeated for ten times. The average of results is presented in Table 12. According to Table 12, the proposed method obtained an average accuracy of 93.70%, an average sensitivity of 90.69%, and an average specificity of 95.82%. In sum, it can be concluded that the proposed OSA detection technique yields better performance than prior works in the literature.

Table 12
Performance of the proposed method on UCDDB database.

Measure	Value
Accuracy	93.70%
Sensitivity	90.69%
Specificity	95.82%

In summary, based on the results, it can be concluded that the combination of CNN and LSTM methods can improve the accuracy of apnea diagnosis. In contrast to the traditional hand-crafted feature extraction methods, the performance of our proposed technique is not dependent on the experts' knowledge from different physiological signals and it is robust. The proposed method can be used in the following practical applications:

- Development of portable home systems for diagnosing sleep apnea
- Development of mobile-based apnea diagnosis systems
- Also, the proposed structure (with some modifications) can be used in other practical applications of physiological signal processing (such as EEG signal processing for seizure detection, ECG signal processing for arrhythmia diagnosis).

4.3. Limitations

Tunning the initial weights for training different deep-learning models affects the performance of models, which is called transfer learning (TL). The main advantage of TL technique is the reduction of learning time and also, making the network with better performance, eliminating the need for a large data, and also avoiding the overfitting problem [64]. Therefore, in future studies, we will investigate the effect of using transfer learning in OSA detection. Moreover, even though the physiological sources of OSA and arrythmias are different, both of them produce similar symptoms to some extent. Both the OSA and arrhythmias cause some changes in the ECG signals. Therefore, development of an OSA detection algorithm (using the ECG signal) in the presence of arrhythmias is essential. So, the next step is to investigate the developing an OSA detection algorithm in the presence of the arrhythmias.

5. Conclusion

In this study, a new algorithm was developed to classify apnea events and normal segments using deep learning structures. The proposed LSTM-CNN model effectively learned features from training data to discriminate apnea events from healthy ones. We have shown the reliability of the proposed method by employing the Apnea-ECG dataset. The most important advantage of the introduced technique is that it solves the need for the feature extraction/selection and classification algorithm. The proposed method provided the average accuracies of 97.21% and 100% for per segment and per-recording classification (on Apnea-ECG dataset), respectively. Also, our model achieves an accuracy of 93.70%, sensitivity of 90.69%, and specificity of 95.82% for UCDDB dataset. The achieved results showed that if the combination of LSTM-CNN is used instead of the conventional feature engineering and classification methods, classification results can be improved significantly. The results based on the Apnea-ECG and UCDDB datasets suggest that the presented algorithm provides significant improvements over the state-of-the-art techniques. Therefore, it can be concluded that the use of deep-learning based technique to diagnose apnea patients would help physicians to make more accurate decisions.

CRediT authorship contribution statement

Asghar Zarei: Conceptualization, Methodology, Data curation, Writing - original draft, Validation. **Hossein Beheshti:** Methodology, Software, Writing - original draft, Investigation. **Babak Mohammadzadeh Asl:** Writing - review & editing, Methodology, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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